

Arvind Yogesh Suresh Babu

savyo@umich.edu | github.com/arvindxyogesh | linkedin.com/in/arvindxyogesh

Summary

MS Data Science candidate (Dec 2026) experienced building and validating ML and data systems for operations, planning, and analytics. I design ETL pipelines, feature-engineering workflows, and evaluation/QA checks; build end-to-end models in PyTorch/TensorFlow; and deploy reproducible services on cloud platforms (AWS/Azure). My GitHub work demonstrates robot analytics, forecasting, model serving, and testing-oriented engineering artifacts.

Skills

Core ML: PyTorch, TensorFlow, scikit-learn, model evaluation, hyperparameter tuning | **GenAI & NLP:** LLMs (fine-tuning & prompt engineering), RAG pipelines, conversational AI, speech/voice processing | **Engineering & Deployment:** Model serving, API design, batching, Docker, CI/CD, AWS, Azure, monitoring | **Testing & QA:** Test planning, test scripts, test data sets, black-box testing, equivalence-class style validation, regression checks, root-cause analysis | **Data & Pipelines:** PySpark, Databricks, dbt, ETL, data validation, telemetry parsing | **Visualization & Product:** Streamlit, Tableau, Power BI, KPI dashboards, stakeholder reporting | **Collaboration:** Cross-functional design, documentation, reproducible runbooks, agile workflows

Work Experience

M-City, UMTRI, Ann Arbor, MI

September 2025 - Present

Research Associate

- Built Spark/Databricks pipelines with dbt to process IoT and mobility telemetry; added data-quality checks and monitoring to catch failures quickly.
- Designed retrieval-augmented and prompt-engineered GenAI components; integrated model endpoints for internal analyst tools and evaluated outputs for reliability.
- Translated product questions into predictive models and dashboards for fleet and sensor analytics; documented schemas, runbooks, and validation steps for reproducibility.

Vestas Wind Technology, Chennai, India

June 2024 - July 2025

Data Science Intern, Mains Supply Planning (SSCBP)

- Supported gearbox supply planning and shipped production forecasting workflows for 200+ operational SKUs; built ETL pipelines with Spark/Kafka to process high-volume telemetry and integrated outputs into planning dashboards.
- Partnered closely with quality assurance teams to design test conditions, test scripts, and test data sets for preprocessing, schema consistency, and forecast outputs; added checks to catch regressions early and reduce manual review effort before release.
- Enhanced reporting infrastructure and built automated pipelines that combined classical forecasting with lightweight GenAI summarization to surface anomalies and suggested actions, reducing manual triage time by 26 hours/week.
- Improved supply visibility and inventory decisions, reducing stockouts by 17%, while designing experiments, model-validation pipelines, and control checks to quantify forecasting performance and business impact.

MES section, IIT Madras, Chennai, India

June 2023 - Aug 2023

Research Intern

- Processed multi-channel sensor data from a 6-axis industrial robot for drilling quality analysis; engineered statistical and spectral features with Butterworth/notch filtering to support behavior and process assessment.
- Developed a computer vision tool for tool tracking via contour detection and optical flow; integrated model-serving endpoints to allow on-demand inference and feedback loops during experiments.
- Created reusable Python scripts for data acquisition and preprocessing and experimented with small CNNs to detect tool anomalies; these artifacts improved reproducibility and reduced manual effort by 20%.

Education

University of Michigan, Ann Arbor, MI Dec 2026

MS Data Science

Relevant Subjects: Applied Machine Learning, Natural Language Processing, Regression Analysis, Advanced DBMS, Advanced Probability and Distributions, Statistical Inference

Anna University, Chennai, India June 2024

BE Mechanical Engineering

Relevant Subjects: C Programming, Python Programming, Data Structures & Algorithms, Robotics Simulations, Control Systems

Certifications & Award

Microsoft: Foundations of AI and Machine Learning Infrastructure (Aug 2025, Credential ID: Q6D72J7U7V2J) |

Google: Advanced Data Analytics Specialization (Oct 2024, Credential ID: BQZ2YSKNFXVO) |

Indian Patent Office: Registered design patent (Jul 2024, Credential ID: 404013-001) for a terrain-adaptable AGV rover architecture.

Project Work

- ML Pipeline Framework:** Built reproducible, production-quality pipeline infrastructure with schema validation, train/test splitting, evaluation harnesses, and standardized reports; includes model training, CI, and deployment patterns.
- PyTorch FastAPI Inference:** Shipped model-serving endpoints and API interfaces to connect ML logic to applications, with batching, error handling, and integration-focused design.
- RAG Document Search:** Built a retrieval-augmented AI agent for unstructured information; engineered vector stores, prompt templates, and evaluation harnesses to ensure grounded responses and reliable behavior under diverse queries.